Lets revise some formulae:

a, b

POR = ab ✓

SOR = a+b ✓

$a^n \pm b^n$

$$a^2 + b^2 = (a+b)^2 - 2ab$$

$$a^3 + b^3 = (a+b)^3 - 3ab(a+b)$$

$$a^4 + b^4 = (a^2+b^2)^2 - 2a^2b^2$$

$$a^5 + b^5 = (a^2+b^2)(a^3+b^3) - a^2b^2(a+b)$$

$$a^6 + b^6 = (a^2+b^2)^3 - 3a^2b^2(a^2+b^2) = (a^3+b^3)^2 - 2a^3b^3$$

$$a^7 + b^7 = (a^3+b^3)(a^4+b^4) - a^3b^3(a+b)$$

$$a^8 + b^8 = (a^4+b^4)^2 - 2a^4b^4$$

coaching center

If 'a' and 'b' are the roots of the equation $x^2 - 14x + 11 = 0$, find $a^2 + b^2$?

अगर a और b समीकरण $x^2 - 14x + 11 = 0$ के मूल हैं तो $a^2 + b^2$ ज्ञात करें|

a) 174 b) 196 c) 218 d) None

$a+b = 14$

$ab = 11$

$= (14)^2 - 2 \times 11$

$= 196 - 22$

$= 174$

coaching center

$\alpha+\beta=6$

$\alpha\beta=6$

If α and β are the roots of the equation $x^2-6x+6=0$, then what is $\alpha^3+\beta^3+\alpha^2+\beta^2+\alpha+\beta$ equal to ?

यदि α और β समीकरण $x^2-6x+6=0$ के मूल हैं, तो $\alpha^3+\beta^3+\alpha^2+\beta^2+\alpha+\beta$ किसके बराबर है ?

a) 150 b) 138 c) 128 d) 228

$= 216 - 3\cdot6\cdot6 + 36 - 2\cdot6 + 6$ (108, 12)

$= 258 - 120 = 138$

coaching center

HW

a and b are the roots of the quadratic equation $x^2 - x - 1 = 0$ what is the value of $a^8 + b^8$?

a तथा b द्विघात समीकरण $x^2 - x - 1 = 0$ के मूल है| $a^8 + b^8$ का मान क्या है?

a) 47 b) 54 c) 59 d) 68

$a+b = 1$, $ab = -1$

$a^2+b^2 = 1^2 - 2\times -1 = 3$

$a^4+b^4 = 3^2 - 2\cdot 1 = 7$

$a^8+b^8 = 7^2 - 2\cdot 1 = 47$

coaching center

$\alpha+\beta=\frac{-b}{a}$

$\alpha\beta=\frac{c}{a}$

If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then the value of $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ is:

अगर समीकरण $ax^2 + bx + c = 0$ के मूल α और β हैं तो $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ का मान ज्ञात करें|

a) $\frac{3bc-a^3}{b^2c}$ b) $\frac{3abc-b^3}{a^2c}$ c) $\frac{3abc+b^3}{a^2c}$ d) $\frac{ab-b^2c}{2b^2c}$

$$\frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{\frac{-b^3}{a^3} + 3\frac{c}{a}\frac{b}{a}\frac{a}{a}}{\frac{c}{a}} = \frac{3abc-b^3}{\alpha^2 \cdot \frac{c}{\alpha}}$$

coaching center

$x^2 - SoRx + PoR = 0$

a^3+b^3 $\quad$ a^3b^3

(-2) $\quad$ (1)

Stress

$a+b=1$

$ab=1$

शांति

$a^3+b^3 = 1-3(1)(1) = -2$

$a^3b^3 = 1$

If a and b are the roots of equation $x^2 - x + 1 = 0$, then which equation will have roots a^3 and b^3?

यदि a तथा b समीकरण $x^2 - x + 1 = 0$ के मूल है, तो किसी समीकरण के मूल a^3 तथा b^3 होंगे?

a) $x^2 + 2x + 1 = 0$

b) $x^2 - 2x - 1 = 0$

c) $x^2 + 3x - 1 = 0$

d) $x^2 - 3x + 1 = 0$

coaching center

$ax^2+bx^2+c=0$

SOR ✗ POR ✓

$POR = -9 \times -1 = 9$ < 9, 1

$SOR = 8+2 = 10$

$x^2-10x+9=0$

In solving a problem, one student makes a mistake in the coefficient of the first degree term and obtains -9 and -1 for the roots. Another student makes a mistake in the constant term of the equation and obtains 8 and 2 for the roots. The correct equation was

एक सवाल को हल करने में, एक छात्र एक घात के पद के गुणांक में गलती करता है और मूल -9 और -1 प्राप्त करता है। एक अन्य छात्र की समीकरण अचर पद में गलती होती है और मूल 8 और 2 प्राप्त करता है। सही समीकरण था ?

a) $x^2 + 10x + 9 = 0$

b) $x^2 + 10x - 9 = 0$

c) $x^2 - 10x + 9 = 0$

d) None of these

coaching center

If the equations $x^2 + 5x + 6 = 0$ and $x^2 + kx + 1 = 0$ have a common root, then what is the value of k ?

यदि समीकरण $x^2 + 5x + 6 = 0$ और $x^2 + kx + 1 = 0$ का एक समान मूल है, तो k का मान क्या होगा?

a) $-\frac{5}{2}$ or $-\frac{10}{3}$ b) $\frac{5}{2}$ or $\frac{10}{3}$ c) $\frac{5}{2}$ or $-\frac{10}{3}$ d) $-\frac{5}{2}$ or $\frac{10}{3}$

POR = 6 < -3, -2

SOR = -5

POR = 1 → -3, $-\frac{1}{3}$; -2, $-\frac{1}{2}$

SOR = -k

$-\left(-2-\frac{1}{2}\right)$ OR $-\left(-3-\frac{1}{3}\right)$ = −SOR = k

$\frac{5}{2}$ OR $\frac{10}{3}$

coaching center

If $sin\theta$ and $cos\theta$ are the roots of the equation $ax^2 - bx + c = 0$, then which one of the following is correct ?

यदि $sin\theta$ और $cos\theta$ समीकरण $ax^2 - bx + c = 0$ के मूल हैं, तो निम्न में से कौन –सा सही है ?

a) $a^2 + b^2 + 2ac = 0$

b) $a^2 - b^2 + 2ac = 0$

c) $a^2 + c^2 + 2ab = 0$

d) $a^2 - b^2 - 2ac = 0$

$\sin\theta + \cos\theta = \frac{b}{a}$

$\sin\theta \cdot \cos\theta = \frac{c}{a}$

$\sin^2\theta + \cos^2\theta = 1$

$\frac{b^2}{a^2} - 2\frac{c}{a} = 1$

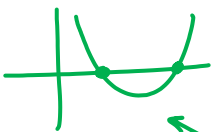
$\frac{b^2 - 2ca}{a^2} = 1$

coaching center

Nature of roots:

$a, b, c \in I$

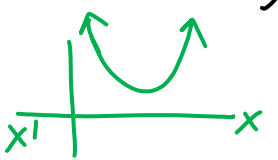
$ax^2+bx+c=0$ $\quad x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$



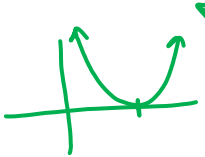
i) If $b^2-4ac>0$ OR $b^2>4ac$ [Real & Unequal]

→ Perfect Square → Rational

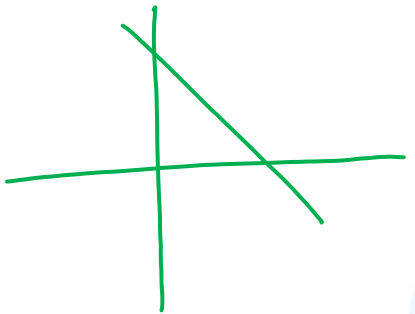
→ X " " → Irrational — Conjugate pairs



ii) If $b^2-4ac<0$ | $b^2<4ac$ [Imaginary & Unequal]



iii) If $b^2-4ac=0$ | $b^2=4ac$ [Real, equal]

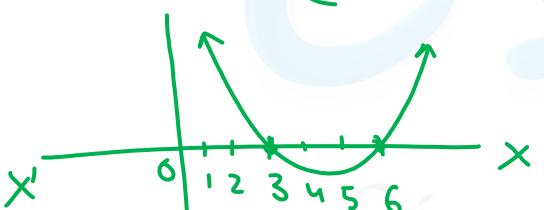


Conjugate Pair

$ax^2+bx+c=0$

$3-\sqrt{5}$, $3+\sqrt{5}$ → if $a,b,c \in$ Rational

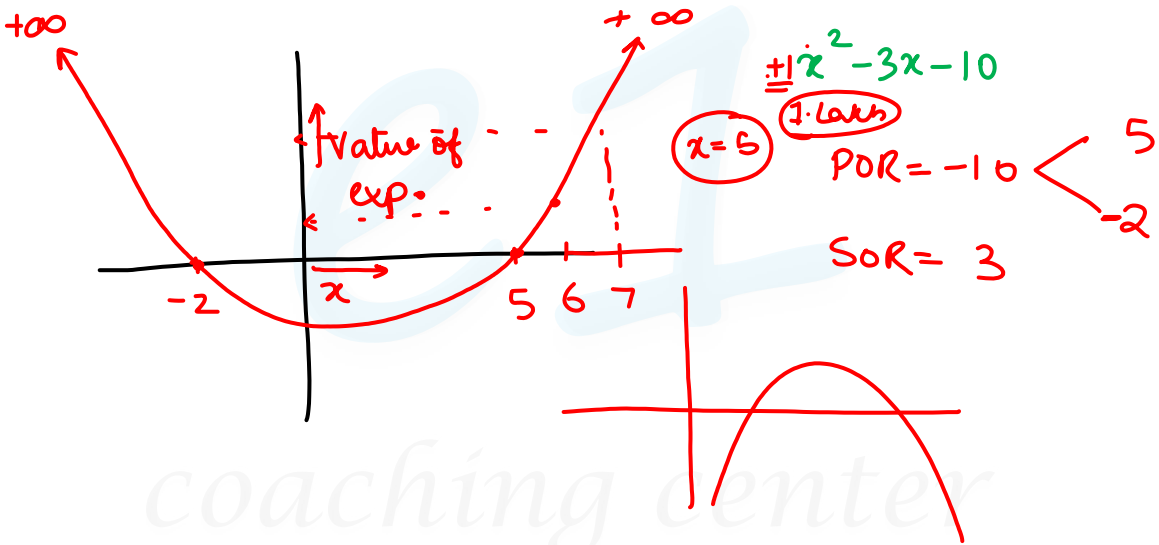
$2+3i$, $2-3i$ → if $a,b,c \in$ Real



$x^2-9x+18=0$

coaching center

Conjugate pairs (संयुग्मी जोड़े):



coaching center

Maximum / minimum value:

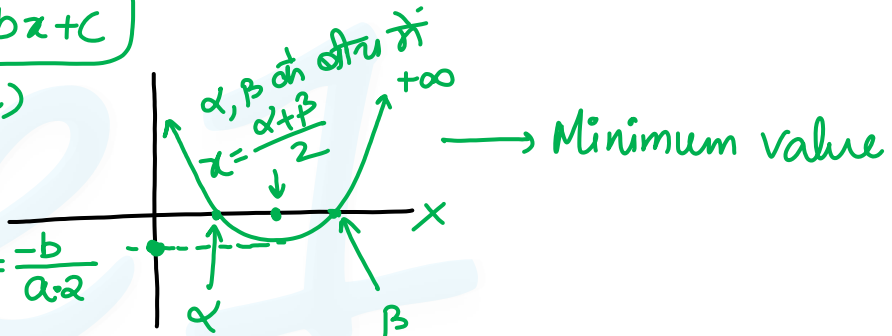
$+3x^2+8x-10$

$x=\frac{-8}{3\times2}$

$x=5$
$x=7$
$x=-8$
$x=-9$

ax^2+bx+c

i) a (+ve)



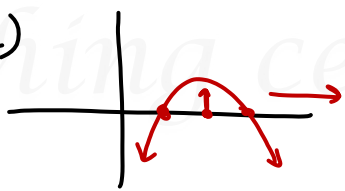
→ Minimum value

$x=\frac{SOR}{2}=\frac{-b}{a\cdot2}$

at $x=-\frac{b}{2a}$

$\text{min value}=\frac{4ac-b^2}{4a}$

ii) a (-ve)



→ max value

Same Concept

coaching center

At what value of p, the expression $p^2 - 4p + 5$ attains its minimum?

P के किस मान के लिए $p^2 - 4p + 5$ का मान न्यूनतम होगा?

a) 2 b) 3 c) 4 d) – 2

at $P = \frac{SOR}{2} = \frac{4}{2} = 2$

Min value-

$4 - 8 + 5 = 1$

$\frac{4ac - b^2}{4a}$

e1 coaching center

At what value of x, does the expression $2x^2 - 48x + 3$ attains it minimum?

x के किस मान के लिए $2x^2 - 48x + 3$ का मान न्यूनतम होगा?

a) 24 b) 12 c) 6 d) – 12

at $x = \frac{48}{2 \cdot 2} = 12$

e1 coaching center

① The expression $(x-3)(x-7)$ attains its minimum value when $x =$

X के किस मान पर $(X-3)(X-7)$ का न्यूनतम मान होगा?

a) 5 b) -5 c) 10 d) -15

$(x-3)(x-7)=0$

$x-3=0 \rightarrow x=3$

$x-7=0 \rightarrow x=7$

$(x+3)(x-2)=0$

$x=-3,2$

at $x=\frac{3+7}{2}=5$

coaching center

② Find the minimum possible value of $2x^2 - 3x + 7$?

$2x^2 - 3x + 7$ का न्यूनतम संभव मान ज्ञात करें|

a) $\frac{47}{8}$ b) $\frac{3}{2}$ c) $\frac{3}{4}$ d) $\frac{3}{8}$

$\frac{4ac-b^2}{4a}$ OR at $x = \frac{3}{2 \cdot 2} = \frac{3}{4}$

$\frac{42 \cdot 7 - 9}{8} = \frac{47}{8}$

coaching center

③ What maximum value can be attained by the expression $-x^2 + 7x - 12$, where $x \epsilon R$.

$-X^2 + 7X - 12$ का अधिकतम संभव मान क्या होगा, अगर X एक वास्तविक संख्या है?

a) 4 b) -4 c) $\frac{1}{4}$ d) $-\frac{1}{4}$

$$\frac{4ac-b^2}{4a} = \frac{4x-1x-12-49}{-4} = \frac{1}{4}$$

coaching center

④ Find the maximum value of the expression $\frac{1}{x^2+5x+10}$.

$\left(\frac{1}{x^2+5x+10}\right)_{max}$ व्यंजक का अधिकतम मान ज्ञात करो |

a) $\frac{15}{2}$ b) 1 c) $\frac{4}{15}$ d) 2

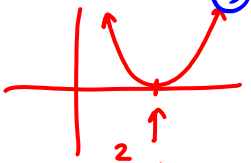
$\frac{4}{15} = \frac{1}{15/4}$

at min of $x^2+5x+10$

$\frac{15}{4} = \frac{4 \cdot 1 \cdot 10 - 25}{4}$

$\left(\frac{1}{x}\right)_{max}$ at x_{min}

coaching center



$b^2-4ac=0$
OR
$b^2=4ac$

⑤ For what value of 'k', does the equation $x^2 + kx + 125 = 100$ has unique positive root?

K के किस मान के लिए समीकरण $x^2 + kx + 125 = 100$ का एकमात्र धनात्मक हल होगा?

$x^2+kx+25=0$

a) 10 b) – 10 c) 10 or – 10 d) None

2017

$k^2 = 4 \cdot 1 \cdot 25$

$k^2 = 100$

$k = \pm 10$

-10 → SOR 10

$+10$ → -10

POR $25 < \begin{matrix} 5 \\ 5 \end{matrix}$ ✓

$25 < \begin{matrix} -5 \\ -5 \end{matrix}$ ✗

coaching center

⑥ What is the maximum integer that 'k' can take such that the roots of the equation $2x^2 + 5x + k$ are real?

अगर समीकरण $2x^2 + 5x + k$ के मूल वास्तविक हैं तो k का अधिकतम पूर्णांक मान ज्ञात करें|

a) 3 b) 4 c) – 3 d) 0

$b^2 > 4ac$

$25 > 4 \cdot 2 \cdot k$

$\Rightarrow k < \frac{25}{8}$

$k < 3\frac{1}{4}$

③

coaching center

(7) What is the smallest integer value of 'k' such that the roots of the equation $x^2 - 5x + (k + 2) = 0$ are imaginary?

अगर समीकरण $x^2 - 5x + k + 2 = 0$ के मूल काल्पनिक हैं तो k का न्यूनतम पूर्णांक मान ज्ञात करें|

a) – 5 b) 4 c) 5 d) none

$b^2 < 4ac$

$25 < 4 \cdot 1 \cdot (k+2)$

$\Rightarrow 4k+8 > 25$

$\Rightarrow k > \frac{17}{4} \mid 4\frac{1}{4}$

coaching center

③ A root of equation $ax^2 + bx + c = 0$ (where a, b and c are rational numbers) is $5 + 3\sqrt{3}$. What is the value of $\frac{a^2+b^2+c^2}{ab+bc+ca}$?

$5-3\sqrt{3}$

समीकरण $ax^2 + bx + c = 0$ (जहाँ a, b तथा c परिमेय संख्याएं है) का मूल $5 + 3\sqrt{3}$ है | तो $\frac{a^2+b^2+c^2}{ab+bc+ca}$ का मान क्या है?

a) $\frac{35}{3}$ b) $\frac{37}{3}$ c) $-\frac{105}{8}$ d) $\frac{105}{8}$

SOR = 10

POR = 25 − 27 = −2

$x^2 - 10x - 2 = 0$

$ax^2 + bx + c = 0$

1 −10 −2

$\frac{1+100+4}{-10+20-2} = \frac{105}{8}$

(9) If $x = \cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{\ldots\infty}}}}}$ then what is the value of x ?

यदि $x = \cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{\ldots\infty}}}}}$ है, तो x का मान क्या है ?

a) $-2 \pm 2\sqrt{3}$

b) Both c & d

c) $\sqrt{3}-1$

d) $-\sqrt{3}-1$

$x = \cfrac{2}{2+x}$

$\Rightarrow 1x^2+2x-2=0$

SOR $= -2$ ✓

POR $= -2$ ✓

$x = \cfrac{-b \pm \sqrt{b^2-4ac}}{2a}$

$\sqrt{3} = 1.732$

$-(\sqrt{3}+1)$

$= -(3-1) = -2$

-ve / +ve

coaching center

Factor/remainder theorem:

$f(x) = ax^4 + bx^3 + cx^2 + dx + e$

$x^3 - 2x^2 - 5x + 6$

$x = -a$

$x + a = 0$

$x = -a$

0

Remainder

$(x+a) \times [\quad] + \text{extra}$

$(x+a)$

$x + a = 0$

$x = -a$

Remainder = $f(-a)$

$\frac{37}{4} = 9 \times 4 + 1$

$\left|\frac{21}{5}\right| = \frac{5 \times 4 + 1}{5}$ (R)

$\frac{20}{5} \Rightarrow 5 \times 4 + 0$ (R)

coaching center

(10) When $f(x) = 15x^3 - 14x^2 - 4x + 10$ is divided by $(3x + 2)$, then the remainder is:

जब $f(x) = 15x^3 - 14x^2 - 4x + 10$ को $(3x + 2)$, से विभाजित किया जाता है, तो _____ शेषफल प्राप्त होता है|

a) 2 b) -1 c) -2 d) 1

$3x+2=0$

$x=-\frac{2}{3}$

$15x^3 - 14x^2 - 4x + 10$

$= -40 - 56 + 24 + 90$ / 9

$= -96 + 114$ / 9 $= \frac{18}{9} = 2$

coaching center

(11) If $px^3 + x^2 + 3x + q$ is exactly divisible by $(x + 2)$ and $(x - 2)$, then the value of p and q are:

यदि $px^3 + x^2 + 3x + q$, $(x + 2)$ एवं $(x - 2)$ से पूर्णत: विभाज्य है, तो p और q के मान _____ है|

a) $p = -\frac{3}{4}$ and $q = 4$ b) $p = \frac{3}{4}$ and $q = 4$

c) $p = \frac{3}{4}$ and $q = -4$ d) $p = -\frac{3}{4}$ and $q = -4$

$-2 = x \leftarrow 0$

$2 = x$

$-8p + 4 - 6 + q = 0$

$8p + 4 + 6 + q = 0$

$-8p + q = 2$

$8p + q = -10$

$p = -\frac{6}{8} = -\frac{3}{4}$ $q = -4$

coaching center

HW

12 If $(x+2)$ and $(x-1)$ are the factors of (x^3+10x^2+mx+n), the values of m and n are:

यदि $(x+2)$ और $(x-1)$, (x^3+10x^2+mx+n) के गुणनखंड हैं, तो m और n का मान ज्ञात करो |

a) $m=5, n=-3$ b) $m=17, n=-8$

c) $m=7, n=-18$ d) $m=23, n=19$

$x+2=0 \Rightarrow x=-2$

$f(-2) \rightarrow -8+40-2m+n=0 \Rightarrow -2m+n=-32$ — ①

$x-1=0 \Rightarrow x=1$

$f(1) \rightarrow 1+10+m+n=0 \Rightarrow m+n=-11$ — ②

② − ①

$3m=21 \Rightarrow m=7$

$n=-18$ in ②

coaching center

(13) If $x^3 + 5x^2 + 10k$ leaves remainder $-2x$ when divided by $x^2 + 2$, then the value of k is:

यदि $x^3 + 5x^2 + 10k$ को $x^2 + 2$ से भाग करें तो, $-2x$ शेष रह जाता है तो k का मान क्या होगा:

a) -2 b) -1 c) 1 d) 2

$x^2 = -2$

$-2x + 5x - 2 + 10k = -2x$

$\Rightarrow 10k = 10$

$k = 1$

coaching center

(14) If $x^{40} + 30$ is divided by $x^4 + 1$, find the remainder

यदि $x^{40} + 30$ को $x^4 + 1$ से विभाजित किया जाए, तो शेषफल ज्ञात कीजिए |

a) 30 b) 31 c) 29 d) 0

$x^4 = -1$

$(x^4)^{10} + 30$

$= (-1)^{10} + 30 = 31$

coaching center

HW (5) If $(x+1)$ and $(x-1)$ are factor of ax^3+bx^2+3x+5. Find the value of $a-2b$.

यदि $(x+1)$ और $(x-1)$, ax^3+bx^2+3x+5 के गुणनखंड है | $a-2b$ का मान ज्ञात करो |

a) 1 b) -13 ~~c) 7~~ d) 0

$x+1=0 \Rightarrow x=-1 \Rightarrow f(-1)=0$

$\Rightarrow -a+b-3+5=0 \Rightarrow -a+b=-2$

$x-1=0 \Rightarrow x=1 \Rightarrow f(1)=0$

$\Rightarrow a+b+3+5=0 \Rightarrow a+b=-8$

$\Rightarrow b=-5$

$\Rightarrow a=-3$

$a-2b=-3+10=7$

coaching center

$x-1=0 \Rightarrow x=1 \Rightarrow f(1)=5$

$\Rightarrow 1-2+3-a+b=5$

$\Rightarrow -a+b=3 \quad -(1)$

$x+1=0 \Rightarrow x=-1 \Rightarrow f(-1)=19$

$1+2+3+a+b=19$

$\Rightarrow a+b=13 \quad -(2)$

$(1)+(2)$ gives $b=8$

$\Rightarrow a=5$

HW

16

If $x^4-2x^3+3x^2-ax+b$ is divided by $x-1$ and $x+1$ the remainders are 5 and 19 respectively. Find the remainder when this polynomial is divided by $x-2$.

यदि $x^4-2x^3+3x^2-ax+b$ को $x-1$ और $x+1$ द्वारा विभाजित किया जाता है तो शेषफल क्रमशः 5 और 19 बचता है | जब इस बहुपद को $x-2$ के द्वारा विभाजित किया जाता है तो शेषफल ज्ञात करो |

a) 10 b) 12 c) -1 d) 8

$x-2=0 \Rightarrow x=2$

$f(2) = 16-16+12-10+8$

$= 10$

coaching center

Higher degree equations:

① $ax^2+bx+c=0$ (α, β)

$\alpha+\beta=-\frac{b}{a}$, $\alpha\beta=\frac{c}{a}$

② $ax^3+bx^2+cx+d=0$ (α, β, γ)

$\alpha+\beta+\gamma=-\frac{b}{a}$

$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{c}{a}$

$\alpha\beta\gamma=-\frac{d}{a}$

coaching center

Representation on graph:

④ $ax^4+bx^3+cx^2+dx+e=0$ $(\alpha, \beta, \gamma, \delta)$

$$\alpha+\beta+\gamma+\delta=\frac{-b}{a}$$

$$\alpha\beta+\beta\gamma+\gamma\delta+\alpha\gamma+\alpha\delta+\beta\delta=\frac{c}{a}$$

$$\alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta=-\frac{d}{a}$$

$$\alpha\beta\gamma\delta=\frac{e}{a}$$

coaching center

Synthetic and long division:

$$x-1 \;\overline{)\; x^3-2x^2-5x+6 \;}\; x^2-x-6$$

$-x^2 \mp x^2$

$-x^2-5x$

$\mp x^2 \pm x$

$-6x+6$

$-6x+6$

X

$x-1=0$

$x=1$

1	1	-2	-5	6
	0	1	-1	-6
	1	-1	-6	0

$x^2-x-6=0$

coaching center

(17) What is $(x-a)(x-b)(x-c)$ equal to?

$(x-a)(x-b)(x-c)$ किसके बराबर है ?

a) $x^3-(a+b+c)x^2+(bc+ca+ab)x-abc$

b) $x^3+(a+b+c)x^2+(bc+ca+ab)x+abc$

c) $x^3-(bc+ca+a)x^2+(a+b+c)x-abc$

d) $x^3+(bc+ca+a)x^2-(a+b+c)x-abc$

$x^2-SOR\,x+POR=0$

$x^2-(a+b)x+ab=0$

a, b, c

$x^3-(SOR)x^2+(SOR_2)x-POR=0$

$x^3-(a+b+c)x^2+(ab+bc+ca)x-abc=0$

coaching center

(13) If α, β and γ are the zeros of the polynomial $f(x) = ax^3 + bx^2 + cx + d$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to

यदि α, β और γ बहुपद $f(x) = ax^3 + bx^2 + cx + d$ के शून्य हैं, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर हैं:

a) $\frac{b^2-ac}{a^2}$ b) $\frac{b^2-2ac}{a}$ c) $\frac{b^2+2ac}{b^2}$ d) $\frac{b^2-2ac}{a^2}$

$$\alpha+\beta+\gamma = \frac{-b}{a}$$

$$\alpha\beta+\beta\gamma+\gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$(\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha)$$

$$= \frac{b^2}{a^2} - 2\cdot\frac{c}{a}\cdot\frac{a}{a}$$

$$= \frac{b^2-2ac}{a^2}$$

coaching center

(19) What are the factors of $x^3 + 4x^2 - 11x - 30$?

$x^3 + 4x^2 - 11x - 30$ के गुणनखंड क्या होंगे ?

a) $(x-2)$, $(x+3)$ and $(x+5)$ b) $(x+2)$, $(x+3)$ and $(x-5)$

c) $(x+2)$, $(x-3)$ and $(x+5)$ d) $(x+2)$, $(x-3)$ and $(x-5)$

$2, -3, -5$

$a+b+c = -4$

$ab+bc+ca = -11$

$abc = 30$

$\pm 6, \pm 5, \pm 1$ ×

$-2, 3, -5$

$(x+2)(x-3)(x+5) = 0$

$-6, -15, +10$

coaching center

(20) If x, y, z are the three factors of $a^3 - 7a - 6$, then the value of x + y + z will be

यदि $a^3 - 7a - 6$ के तीन गुणनखंड x, y, z हैं, तो x + y + z का मान होगा ?

a) 3a b) 3 c) 6 d) a

roots ✗

$a^3 - 0\,a^2 - 7a - 6$

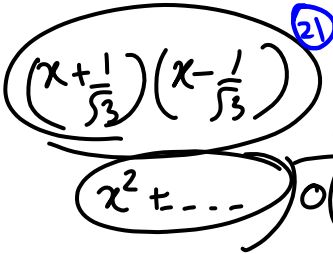
$P, q, r \rightarrow (a-P)_{x}(a-q)_{y}(a-r)_{z}$

$= 3a - (P+q+r)$

$P+q+r = \frac{0}{1} = 0$

$= 3a - 0 = 3a$

coaching center



2) Find the rational root of polynomial $f(x) = 3x^4 - 15x^3 + 17x^2 + 5x - 6$ if one roots is $-\frac{1}{\sqrt{3}}$.

बहुपद $f(x) = 3x^4 - 15x^3 + 17x^2 + 5x - 6$ का परिमेय मूल क्या होगा। यदि एक मूल $-\frac{1}{\sqrt{3}}$ है।

a) -3, 2 b) 3, 2 c) 3,-2 d) -3, -2

$\frac{0}{a+b} + \frac{5}{c+d} = \frac{15}{3} = 5 \rightarrow c+d = 5$

$\frac{-\frac{1}{3}}{abcd} = \frac{-6}{3} = -2 \rightarrow cd = -2 \times -3 = 6 < \begin{matrix} 2 \\ 3 \end{matrix}$

$-\frac{1}{\sqrt{3}}, +\frac{1}{\sqrt{3}}$, ,

a, b, c, d

2, 3

① Video

② PDF → Solve

PDF + Mock